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PROJECT REPORT

A Report submitted in partial fulfilment of the requirements for the award of degree of

BACHELORS OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING **MACHINE LEARNING PROJECT**

Comparative Analysis of Machine Learning Algorithms for Customer Churn Prediction

Submitted By:

Ritesh Chourasia (CSJMA23001390323)

Pawan Kumar (CSJMA23001390316)

Rajdeep Ranjan (CSJMA23001390321)

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Submitted To:

Dr. Alok Kumar

Department of Computer Science & Engineering

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Submitted By:

- Ritesh Chourasia
- Pawan Kumar
- Rajdeep Ranjan

2.Abstract

Customer churn prediction is one of the most important applications of machine learning in the telecom industry. Telecom companies face major financial losses when customers discontinue their services. Predicting churn in advance helps companies improve customer retention strategies and reduce revenue loss.

In this project, the IBM Telco Customer Churn Dataset was used to analyze customer behavior and predict whether a customer is likely to churn or not. Multiple machine learning algorithms including Logistic Regression, Naive Bayes, Decision Tree, K-Nearest Neighbors (KNN), Random Forest, Support Vector Machine (SVM), and XGBoost were implemented and compared on the same dataset.

The project involved complete preprocessing of categorical and numerical data using pipelines and column transformers. Exploratory Data Analysis (EDA) was performed to identify important patterns and correlations related to customer churn. Hyperparameter tuning and ensemble learning techniques were also applied to improve model performance.

The best-performing model was selected based on evaluation metrics such as Accuracy, Precision, Recall, and F1-Score. The final trained model was exported and prepared for deployment in a web-based application

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3. Literature Survey

Customer churn prediction has become a major area of research in machine learning and data analytics. Companies in the telecom industry continuously analyze customer data to understand customer behavior and prevent customer loss.

Traditional statistical approaches were initially used for churn prediction, but modern machine learning algorithms have significantly improved prediction accuracy. Logistic Regression and Decision Trees are among the earliest methods used for churn analysis because of their simplicity and interpretability.

Ensemble learning techniques such as Random Forest and Gradient Boosting later became popular because they reduce overfitting and improve prediction performance. Advanced boosting algorithms such as XGBoost have shown excellent performance on structured datasets due to their optimization and regularization techniques.

Several studies also highlight the importance of preprocessing techniques such as feature encoding, scaling, missing value handling, and feature engineering in improving model performance.

In this project, multiple machine learning algorithms were implemented and comparatively analyzed to identify the most suitable model for telecom customer churn prediction.

4.Data Collection and Preprocessing

4.1 Dataset Description

The project uses the IBM Telco Customer Churn Dataset. The dataset contains information about telecom customers including demographic information, subscription details, billing information, internet services, and customer churn status.

Dataset Information

- Total Rows: 7043
- Total Columns: 21
- Target Variable: Churn
- Dataset Type: Classification Dataset

Numerical Features

- tenure
- MonthlyCharges
- TotalCharges

Categorical Features

- gender
- Partner
- Dependents
- PhoneService
- InternetService
- Contract
- PaymentMethod
- OnlineSecurity
- StreamingTV
- StreamingMovies
- TechSupport
- etc.

4.2 Data Cleaning

The dataset preprocessing included the following steps:

Removal of Unnecessary Columns

The customerID column was removed because it is only a unique identifier and does not contribute to prediction.

Handling Missing Values

The TotalCharges column contained blank values which were converted into numerical format using pandas. Missing values were filled using the median value of the column.

Feature Encoding

Categorical variables were transformed into numerical format using:

- Binary Encoding
- One Hot Encoding

Feature Scaling

Numerical columns were standardized using StandardScaler to normalize feature distributions.

Pipeline and Column Transformer

A complete preprocessing pipeline was created using:

- Pipeline
- ColumnTransformer
- FunctionTransformer

This ensured consistent preprocessing during model training and deployment.

4.3 Exploratory Data Analysis (EDA)

EDA was performed to understand customer behavior and identify important features influencing churn.

Visualizations Performed

- Churn Distribution Countplot
- Histograms
- Boxplots
- Correlation Analysis
- Feature Correlation Barplots
- Heatmaps
- Pairplots

Important Insights

- Customers with month-to-month contracts showed higher churn rates.
- Customers using fiber optic internet were more likely to churn.

- Long tenure customers had lower churn probability.
- Customers using electronic check payment methods showed higher churn rates.
- Customers with tech support and online security services had lower churn rates.

5. Algorithms Used with Details

5.1 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. It is mainly used when the output has two possible categories such as Yes/No, True/False, or Churn/No Churn. In this project, Logistic Regression is used to predict whether a telecom customer is likely to churn or stay with the company.

The algorithm works using a sigmoid function, which converts values into probabilities between 0 and 1. Based on the probability score, the model predicts the final class. Since customer churn prediction is a binary classification problem, Logistic Regression is considered a suitable algorithm for this project.

Logistic Regression helps in understanding the relationship between customer-related features such as tenure, monthly charges, contract type, and payment method with customer churn behavior. It is simple, fast, and easy to implement, making it one of the most commonly used classification algorithms.

Advantages

- Fast and simple to implement
- Easy to interpret and understand
- Performs well on linearly separable data
- Suitable for binary classification problems

Disadvantages

- Limited for highly nonlinear relationships
- Sensitive to outliers in the dataset
- Performance depends on proper feature engineering and preprocessing

5.2 Naive Bayes

Naive Bayes is a supervised machine learning classification algorithm based on Bayes Theorem. It is mainly used for classification tasks and works by calculating probabilities for different classes. In this project, Naive Bayes is used to predict whether a telecom customer is likely to churn or not based on customer-related features.

The algorithm assumes that all input features are conditionally independent of each other, meaning one feature does not directly affect another. Although this assumption may not always be true in real-world datasets, Naive Bayes still performs efficiently and provides good classification results.

Naive Bayes is especially useful for handling categorical data and high-dimensional datasets. It is computationally fast and requires less training time, making it suitable for customer churn prediction problems.

Advantages

- Fast computation and training speed
- Performs well with categorical data
- Works efficiently on high-dimensional datasets
- Simple and easy to implement

Disadvantages

- Assumption of feature independence may not always hold true
 - Performance may decrease when features are highly related
 - Less effective for complex relationships in data
-

5.3 Decision Tree

Decision Tree is a supervised machine learning algorithm used for both classification and regression problems. It works by splitting the dataset into different branches based on feature conditions, creating a tree-like structure. In this project, Decision Tree is used to predict customer churn by analyzing customer information such as contract type, tenure, internet service, and payment method.

The algorithm makes decisions by selecting the most important features and dividing the data into smaller groups. Each branch in the tree represents a condition, while the final nodes represent the prediction outcome such as churn or no churn. Decision Tree is easy to understand and visualize, making it useful for interpreting prediction results.

One of the major advantages of Decision Tree is that it can handle nonlinear relationships and does not require feature scaling. However, if the tree becomes too complex, it may lead to overfitting and reduce model performance on unseen data.

Advantages

- Easy to interpret and visualize
- Handles nonlinear relationships effectively
- No feature scaling required
- Works with both numerical and categorical data

Disadvantages

- Can overfit easily on training data
 - Sensitive to small changes in data
 - Complex trees may reduce prediction accuracy
-

5.4 K-Nearest Neighbors (KNN)

- K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and prediction tasks. It classifies data points based on the majority class of the nearest neighboring data points. In this project, KNN is used to predict customer churn by comparing customer information with similar existing customers in the dataset.

- The KNN algorithm works by finding the value of **K**, which represents the number of nearest neighbors considered for prediction. It calculates the distance between data points and assigns the class based on the majority category among the nearest neighbors. For example, if most nearby customers belong to the churn category, the new customer is also predicted to churn.
- KNN is simple to implement and does not require a separate training phase. However, it becomes computationally expensive for large datasets and is sensitive to feature scaling, which is why preprocessing and normalization are important.

- **Advantages**

- • Simple and easy to understand algorithm
- • No separate training phase required
- • Works well for classification problems
- • Effective for smaller datasets

- **Disadvantages**

- • Computationally expensive for large datasets
 - • Sensitive to feature scaling
 - • Performance depends on the selection of K value
-

5.5 Random Forest

Random Forest is a supervised machine learning algorithm based on the ensemble learning technique. It combines multiple Decision Trees to improve prediction accuracy and reduce overfitting. In this project, Random Forest is used to predict customer churn by analyzing customer behavior and service-related features.

The algorithm works by creating multiple decision trees using different subsets of the dataset. Each tree makes its own prediction, and the final result is obtained through **majority voting**. This approach improves model stability and provides more accurate predictions compared to a single Decision Tree.

Random Forest performs effectively on large datasets and can handle both numerical and categorical features. It is also useful for identifying important features influencing customer churn. However, due to multiple trees, it requires higher computational power and training time.

Advantages

- Reduces overfitting compared to Decision Tree
- Provides high prediction accuracy
- Handles large datasets effectively
- Works well with both numerical and categorical data

Disadvantages

- More computationally expensive
 - Training time is higher compared to simple models
 - Model interpretation can be difficult
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5.6 Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification problems. It works by creating an optimal **hyperplane** that separates different classes with the maximum possible margin. In this project, SVM is used to classify telecom customers into churn or non-churn categories based on customer-related features.

The main objective of SVM is to find the best boundary that clearly separates different classes in the dataset. It identifies important data points called **support vectors**, which help define the decision boundary. SVM performs well in high-dimensional datasets and provides strong classification performance.

SVM is particularly useful when the dataset contains complex patterns and multiple features. However, it may become slower for large datasets and requires careful tuning of hyperparameters to achieve better performance.

Advantages

- Effective in high-dimensional spaces
- Provides good classification performance
- Works well for binary classification problems
- Handles complex decision boundaries effectively

Disadvantages

- Slower for large datasets
- Sensitive to hyperparameter tuning
- Requires more computational resources compared to simple algorithm

5.7 XGBoost

XGBoost (Extreme Gradient Boosting) is an advanced supervised machine learning algorithm based on the boosting technique. It improves prediction performance by building multiple models sequentially, where each new model attempts to correct the errors made by the previous model. In this project, XGBoost is used for customer churn prediction to improve classification accuracy.

The algorithm works by combining several weak learning models, usually decision trees, to create a strong predictive model. XGBoost is highly efficient and capable of handling complex relationships between features such as customer tenure, internet services, monthly charges, and contract type.

One of the major strengths of XGBoost is that it includes **regularization**, which helps reduce overfitting and improves model performance. It is widely used in machine learning competitions because of its speed and high prediction accuracy. However, the algorithm requires proper parameter tuning and higher computational power.

Advantages

- High prediction performance and accuracy
- Handles complex relationships effectively

- Includes regularization to reduce overfitting
- Works efficiently with large datasets

Disadvantages

- Computationally intensive
- Requires parameter tuning for better performance
- More complex compared to basic machine learning algorithms

6. Implementation Methodology

The implementation process of the customer churn prediction system followed a structured workflow to ensure accurate predictions and proper data handling. Each stage of implementation is explained below:

1. Dataset Collection

The first step involved collecting the IBM Telco Customer Churn Dataset. The dataset contains customer demographic details, subscription information, billing details, internet services, and churn status used for prediction.

2. Data Cleaning

The dataset was cleaned by removing unnecessary columns and handling missing values. The **customerID** column was removed because it does not contribute to prediction, and missing values in **TotalCharges** were handled properly.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand customer behavior and identify important factors affecting churn. Different visualizations such as countplots, histograms, boxplots, and heatmaps were used to analyze patterns in the dataset.

4. Feature Engineering

Feature engineering was applied to improve model performance by selecting useful features and preparing the dataset for machine learning algorithms.

5. Feature Encoding

Categorical features were converted into numerical form using techniques such as **Binary Encoding** and **One Hot Encoding**, allowing machine learning models to process the data effectively.

6. Feature Scaling

Numerical columns were standardized using **StandardScaler** to normalize data values and improve the performance of algorithms sensitive to feature scales.

7. Train-Test Split

The dataset was divided into training and testing sets. The training data was used to train the model, while testing data was used to evaluate prediction performance.

8. Pipeline Creation

A preprocessing pipeline was created using **Pipeline** and **ColumnTransformer** to automate data preprocessing and ensure consistent transformations during training and testing.

9. Model Training

Multiple machine learning algorithms such as Logistic Regression, Decision Tree, Random Forest, SVM, and XGBoost were trained on the processed dataset.

10. Hyperparameter Tuning

Hyperparameter tuning was performed to optimize model performance by selecting the best parameter values for machine learning algorithms.

11. Model Evaluation

The trained models were evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

12. Comparative Analysis

The performance of different algorithms was compared to identify the best-performing model for customer churn prediction.

13. Model Export

The best-performing trained model was saved for future use and deployment using model serialization techniques.

14. Deployment Preparation

The final model was prepared for deployment so that it can be integrated into an application for real-time customer churn prediction.

The project was implemented using Python and machine learning libraries including:

- Pandas – Used for data manipulation and preprocessing
- NumPy – Used for numerical operations
- Matplotlib – Used for data visualization
- Seaborn – Used for statistical plotting and EDA
- Scikit-learn – Used for preprocessing, training, and evaluation
- XGBoost – Used for advanced boosting-based prediction Pipelines and ColumnTransformers were used to automate preprocessing and prevent data leakage.

7. Results and Comparison

The performance of different machine learning algorithms was evaluated to identify the most accurate and reliable model for customer churn prediction. Various evaluation metrics were used to measure prediction performance and compare the effectiveness of each algorithm.

Accuracy: Measures the percentage of correctly predicted instances by the model. Higher accuracy indicates better prediction performance.

- **Precision:** Measures how many predicted churn cases were actually correct. It helps reduce false positive predictions.

- **Recall:** Measures the model's ability to correctly identify actual churn customers. Higher recall means fewer churn customers are missed.

- **F1-Score:** It is the balanced measure of precision and recall, providing an overall performance score.

- **Confusion Matrix:** A matrix used to visualize model predictions by showing correct and incorrect classifications.

Model Comparison Table

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.821150	0.686154	0.597855	0.638968
10	Stacking	0.816182	0.686275	0.563003	0.618557
4	SVM	0.814053	0.700361	0.520107	0.596923
6	AdaBoost	0.813343	0.682119	0.552279	0.610370
9	Voting	0.809794	0.696629	0.498660	0.581250
7	Gradient Boosting	0.809084	0.672185	0.544236	0.601481
3	Random Forest	0.804826	0.682836	0.490617	0.570983
8	Random Forest	0.792051	0.649254	0.466488	0.542902
5	Bagging	0.779986	0.622568	0.428954	0.507937
2	Decision Tree	0.715401	0.462366	0.461126	0.461745
1	Naive Bayes	0.665720	0.435864	0.892761	0.585752

Hyperparameter Tuning

Hyperparameter tuning was performed using:

- GridSearchCV
- Depth tuning for Decision Tree
- K-value tuning for KNN
- Estimator tuning for Random Forest
- Parameter tuning for SVM
- Estimator tuning for XGBoost

Graphs were plotted to analyze the impact of different hyperparameters on model accuracy.

8. Best Model Selection

Because the customer churn dataset was imbalanced, relying only on accuracy was not sufficient for selecting the best-performing model. In an imbalanced dataset, a model may achieve high accuracy by predicting the majority class more frequently while failing to correctly identify actual churn customers. Therefore, greater importance was given to metrics that evaluate minority class prediction performance effectively.

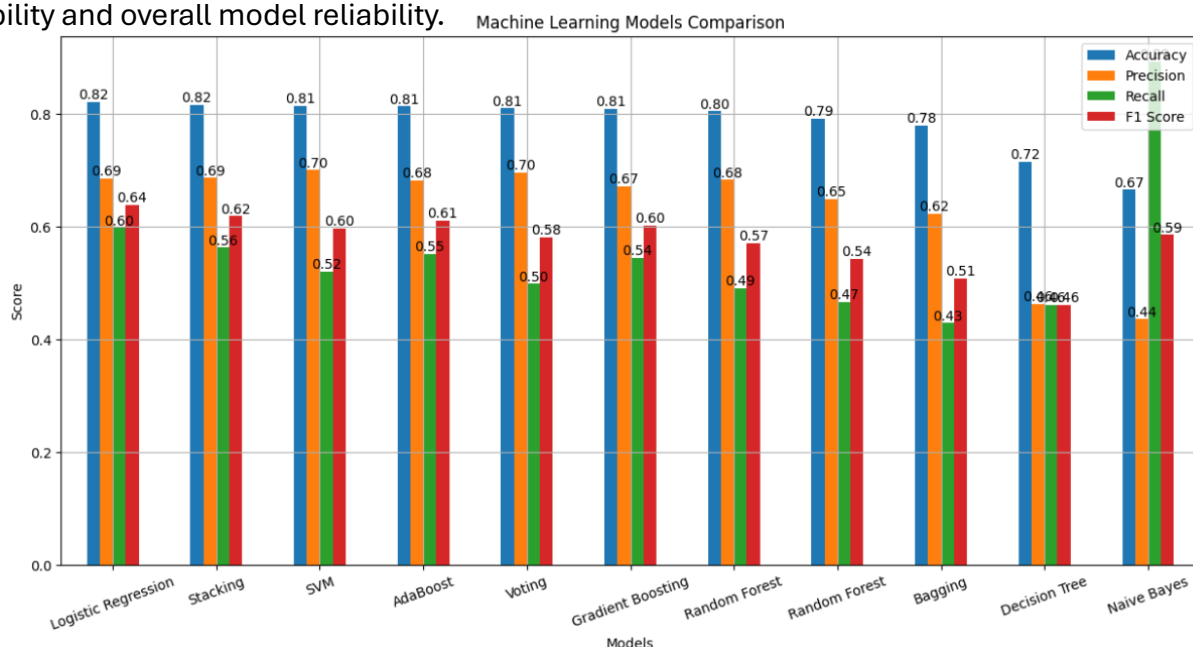
The following criteria were used for selecting the best model:

- **Recall:** Considered the most important metric because the primary objective was to correctly identify customers who are likely to churn. A higher recall reduces false negatives and ensures that fewer actual churn customers are missed.
- **F1-Score:** Used to maintain a balance between precision and recall for overall classification performance.
- **Accuracy:** Considered as a supporting metric to evaluate overall prediction correctness.
- **Generalization Ability:** Used to ensure that the model performs well on unseen data without overfitting.

Several machine learning algorithms were implemented and compared for customer churn prediction. After comparative analysis, Logistic Regression demonstrated the best overall performance for the given dataset. Although some complex models achieved competitive accuracy, Logistic Regression provided better balance between recall, F1-score, stability, and generalization performance.

Logistic Regression performed effectively in identifying churn customers while maintaining consistent prediction behavior on unseen data. Since the dataset was imbalanced, the model's recall performance played a major role in the final selection process. The algorithm also showed lower risk of overfitting and provided interpretable prediction behavior, making it suitable for practical churn analysis.

Based on evaluation metrics and comparative analysis, **Logistic Regression** was selected as the final model for customer churn prediction because it achieved the best balance between churn detection capability and overall model reliability.



9. Deployment Details

The final selected machine learning model was prepared for deployment in a web-based application to enable real-time customer churn prediction. Deployment helps convert the trained model into a practical system where users can input customer details and receive instant prediction results.

The deployment phase focused on creating an interactive and user-friendly interface that allows users to easily access the prediction system.

Deployment Features

The web-based application includes the following features:

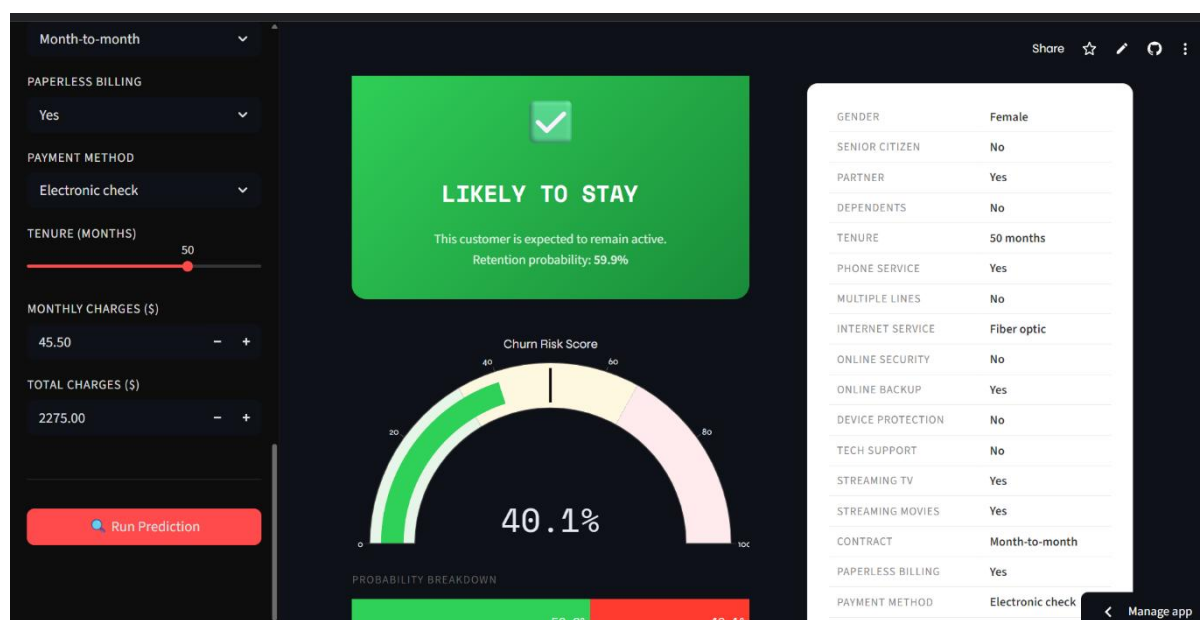
- **User Input Form:** A form was created to allow users to enter customer details such as tenure, contract type, internet service, monthly charges, and payment method.
- **Model Selection Option:** Users can select different machine learning models to compare prediction results and performance.
- **Real-Time Prediction:** The application provides instant churn prediction based on user input data.
- **Churn Probability Prediction:** Along with prediction results, the system displays the probability of customer churn to improve decision-making.

Deployment Tools Used

The deployment process was implemented using the following technologies:

- Flask / FastAPI – Used for backend development and API integration.
- HTML – Used to design the structure of web pages.
- CSS – Used to improve the visual appearance and styling of the application.
- JavaScript – Used to add interactivity and dynamic functionality.
- Joblib – Used to save and load trained machine learning models.

Screenshot:



10. GitHub Repository Link

The complete project source code and implementation files are maintained in a GitHub repository for better accessibility, version control, and project management.

GitHub Repository:

(Add your GitHub repository link here)

The repository contains all important files related to the customer churn prediction project, including source code, datasets, trained models, and deployment files. It helps in maintaining project transparency and allows easy collaboration and future modifications.

Repository Contents

- **Source Code:** Contains Python scripts used for data preprocessing, model training, evaluation, and prediction.
- **Jupyter Notebooks:** Includes notebook files used for exploratory data analysis (EDA), model implementation, and experimentation.
- **Dataset:** Contains the IBM Telco Customer Churn Dataset used for training and testing machine learning models.
- **Trained Models:** Includes saved machine learning models exported in **.pkl** format for future prediction and deployment.
- **Documentation:** Contains project reports, methodology, implementation details, and related documents.
- **Deployment Files:** Includes backend and frontend files required for deploying the machine learning model into a web-based application.

Link: https://github.com/Ritesh-A2/Churn_Analysis_using_multiple_Algorithms

11. Conclusion

This project successfully implemented multiple machine learning algorithms for telecom customer churn prediction. The main objective of the project was to analyze customer behavior and predict whether a customer is likely to churn or continue using telecom services. Different machine learning models were trained and evaluated to identify the best-performing algorithm.

The project highlighted the importance of **data preprocessing, feature engineering, and comparative analysis** in building an effective machine learning system. Proper handling of missing values, feature encoding, scaling, and preprocessing pipelines significantly improved model performance and prediction reliability.

Through **Exploratory Data Analysis (EDA)**, several important business insights related to customer retention were identified. It was observed that customers with month-to-month contracts, fiber optic internet services, and electronic check payment methods showed higher churn rates, while long-tenure customers and customers with additional support services had lower churn probability.

Among all implemented models, ensemble learning algorithms such as **Random Forest** and **XGBoost** achieved better prediction performance compared to traditional machine learning models. Due to its boosting mechanism and sequential error correction capability, **XGBoost was identified as the best-performing model** for customer churn prediction.

The project also demonstrated the practical use of **machine learning pipelines, hyperparameter tuning, model comparison, and deployment preparation**, making the system suitable for real-world applications. Overall, this project provides a useful approach for telecom companies to identify high-risk customers and improve customer retention strategies through data-driven decision-making.

12. Future Scope

The customer churn prediction project can be further improved by integrating advanced technologies and intelligent systems to enhance prediction accuracy and real-time decision-making. Several future improvements can be implemented to make the system more efficient and practical for real-world telecom applications.

The following enhancements can be considered in future development:

- **Deep Learning Models:** Advanced deep learning techniques such as Artificial Neural Networks (ANN) and Deep Neural Networks (DNN) can be implemented to improve prediction accuracy and handle complex customer behavior patterns.
- **Real-Time Churn Prediction Systems:** The model can be integrated with real-time telecom databases to predict customer churn instantly and support proactive customer retention strategies.
- **Cloud Deployment:** The application can be deployed on cloud platforms for better scalability, accessibility, and performance in large-scale environments.
- **Explainable AI Techniques:** Explainable AI methods can be used to understand why the model predicts customer churn, helping businesses make more transparent decisions.
- **SHAP Value Analysis:** SHAP (SHapley Additive exPlanations) can be implemented to identify feature importance and explain how each feature influences prediction results.
- **Live Customer Analytics Dashboards:** Interactive dashboards can be developed to visualize customer behavior, churn trends, and prediction insights in real time.
- **Automated Model Updating:** The system can be enhanced with automation to retrain models automatically using new customer data, ensuring updated and accurate predictions over time.

Overall, future improvements can make the churn prediction system more intelligent, scalable, and suitable for real-world telecom business environments.

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